MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCAQC202 – Anatomy and Physiology of Finfishes and Shellfishes

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer All Questions
Each question carries 1 mark

1. What is Gastric Mill? [1] [U]

2. Name the junction between a.) Pharynx and Oesophagus b.) Stomach and Intestine [1] [R]

3. What is Circulus Cephalicus in Fish Arterial System [1,2] [U]

4. Distinguish between Holobranch and Hemibranch? [1,2] [U]

5. What are Organs of Bojanus? [3] [U]

6. Name two hormones involved in Osmoregulation in Finfishes [3] [R]

7. Write a note on Water Tubes in Bivalves? [4] [U]

8. Name any two dyes used for Marking in fishes? [5] [R]

9. What is Gonadosomatic Index? [5] [U]

10. Name the largest Otolith in fish used for age and growth studies [5] [R]

[1 x 10 = 10] Marks

Part B
Short Answer Questions
Answer any 5 Questions
Each question carries 6 marks

11. Explain the digestive system and method of digestion in Bivalves? [1] [U]

12. Explain Portal Systems in Teleosts with suitable examples? [1,2] [U]

13.Explain different types of Social Aggregation in fishes? [1] [U]

14. Explain the Feeding mechanism in Cephalopods? [4] [U]

15. Describe the structure of nephrons in freshwater teleosts and compare it with that of Marine teleosts [3] [A]

16. Elaborate upon the various types of feeding in fishes with suitable examples [3] [A]

17. Explain Cube Law? Distinguish between Isometric and Allometric growth in fishes? [5] [U]

18. Define size at first maturity? How is it determined? [5] [U]

19. Explain Fecundity and its determination? [5] [U]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer any one Question
Each question carries 10 marks

20. Describe Migration in Fishes with suitable examples [1] [U]

21. Explain Osmoregulatory mechanisms in teleosts and Elasmobranchs [3] [A]

22. Explain the different indirect mechanisms of determination of age and growth in fishes [5] [An]

[10 x 1 = 10] Marks